Version:1

Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

INTRAOPERATIVE ADMINISTRATION OF OBSTETRIC DRUGS

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1. Introduction and Purpose

Obstetricians often request that anaesthetists give certain non-anaesthetic drugs intraoperatively such as antibiotics or drugs that cause contraction or relaxation of the uterus.

2. Scope

This guideline is intended as a reference for trainee anaesthetists seeking advice regarding the administration of such drugs.

3. Points to consider

3.1 **Antibiotic Prophylaxis**

Antibiotic prophylaxis is administered intravenously just prior to the commencement of Caesarean Section, in order to reduce the incidence of wound infection and endometritis.

Cefuroxime (dose = 1.5G iv) and Metronidazole (dose = 500mg iv) are the first line antibiotics.

If the patient has had anaphylaxis to penicillin in the past, they should be given Clindamycin (dose = 600mg iv) and Gentamycin (dose = 120mg iv)

3.2 **Syntocinon bolus**

Indication
Prevention of postpartum haemorrhage due to uterine atony in patients undergoing Caesarean Section

(Give very cautiously in severe cardiovascular disease & severe pre-eclampsia)

Side Effects
Nausea, vomiting, arrhythmia, headache

Dose
5iu by slow intravenous injection after delivery

3.3 **Oxytocin (*Syntocinon*) infusion**

Indication
Postpartum haemorrhage due to uterine atony in patients unresponsive to initial oxytocin bolus

Dose
40iu oxytocin diluted into 500ml 0.9% saline (prepared by midwife for Cat. 1 - 3 LSCS and by the anaesthetist for Cat. 4 LSCS). Infused at 125ml/hr via infusion pump (set up by midwife)

3.4 **Ergometrine**

Indication
Postpartum haemorrhage due to uterine atony in patients unresponsive to oxytocin

Contra-indication
Severe cardiac disease, vascular disease, sepsis, severe hypertension, eclampsia

Side Effects
Nausea, vomiting, headache, dizziness, tinnitus, abdominal pain, chest pain, palpitation, dyspnoea, bradycardia, transient hypertension, vasoconstriction

Dose
250mcg intramuscularly followed by 250mcg (diluted to 5ml with 0.9% saline) slowly intravenously over 3 – 5 minutes

Syntometrine is a commercial mixture of ergometrine 500mcg and oxytocin 5iu in a 1ml vial. The dose is one vial given intramuscularly.

3.5 **Carboprost (Hemabate)**

Indication
Postpartum haemorrhage due to uterine atony in patients unresponsive to ergometrine and oxytocin

Contra-indication
Cardiac, renal, pulmonary or hepatic disease

Side Effects
Nausea, vomiting and diarrhoea, hyperthermia and flushing, bronchospasm

Dose
250mcg intramuscularly (**NOT INTRAVENOUSLY !**) repeated if necessary at intervals of not less than 15 minutes up to a total dose of 2mg.

3.6 **Glyceryl Trinitrate (GTN)**

Indication
When relaxation of the uterus is needed e.g. transverse lie

Contra-indication
Hypovolaemia, hypotension, hypertrophic cardiomyopathy, aortic or mitral stenosis

Side Effects
Headache, dizziness, tachycardia and hypotension

Dose
1- 2 spray doses (0.4 - 0.8mg) sublingually
or
0.15mg - 0.3mg very slowly intravenously over 3 - 5 minutes*

*dilute 1ml of 1mg/ml GTN into 10ml with 0.9% saline thus creating a 0.1mg/ml solution. Give 1.5 – 3ml of this solution slowly iv over 3 – 5 minutes

4. References

1. British Medical Association and Royal Pharmaceutical Society of Great Britain. *British National Formulary* (March 2013) BMJ Group and RPS Publishing

2. Caponas G. Glyceryl trinitrate and acute uterine relaxation : a literature review. *Anaesthesia and Intensive Care*. 2001 Apr; 29(2) : 163 - 77